综合练习

人员

于子珈、牛同泽、杨瑾硕、周子一 到课

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/2901

比赛概况 题目列表 选择题列表 提交记录 实时榜单 选择题排行榜 编辑比赛

王向东老师周日十点半C++对顶堆

刷新

#	用户名	姓名	编程分	时间	A	B	C	D
1	siyunxin	司云心	300	2139	100	100	100	
2	xieyakai	谢亚锴	300	3408	100	100	100	
3	yangjinshuo	杨谨硕	200	1080	100	100		
4	yuzijia1	于子珈	200	1583	100	100		
5	niutongze	牛同泽	200	2156	100	100		
6	donghaozhen	董浩桢	109	263	100	9		
7	lizihan	李子瀚	100	88	100			
8	qinxiansen	秦显森	0	0	0			

本周作业

https://cppoj.kids123code.com/contest/3157 (课上讲了 A ~ C 题, 课后作业是 D 题)

课堂表现

今天题目整体不算难, 同学们课上做题也普遍做的都比较好。

课堂内容

All Assign Point Add

整体修改用 K 变量表示, 单个修改用 map 存储

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

signed main()
{
  int n, K = 0; cin >> n;
  map<int,int> mp;
  for (int i = 1; i <= n; ++i) {
    int x; cin >> x; mp[i] = x;
```

```
  }

  int Q; cin >> Q;
  while (Q -- ) {
    int op; cin >> op;
    if (op == 1) {
      int x; cin >> x; mp.clear(); K = x;
    } else if (op == 2) {
      int x, t; cin >> x >> t;
      if (mp.count(x)) mp[x] += t;
      else mp[x] = K+t;
    } else {
      int x; cin >> x;
      if (mp.count(x)) cout << mp[x] << endl;
      else cout << K << endl;
    }
  }
  return 0;
}
```

Find Permutation

根据大小关系建边, 跑拓扑排序, 看是否存在唯一的拓扑序

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
vector<int> vec[maxn];
int deg[maxn];
int f[maxn];

int main()
{
  int n, m; cin >> n >> m;
  set<pair<int,int>>s;
  while (m -- ) {
    int a, b; cin >> a >> b; s.insert({a,b});
  }
  for (pair<int,int> it : s) {
    int a = it.first, b = it.second;
    vec[a].push_back(b); ++deg[b];
  }

  queue<int> q;
  for (int i = 1; i <= n; ++i) {
    if (!deg[i]) q.push(i);
  }
  int cnt = 0;
  while (!q.empty()) {
```

```
    if ((int)q.size() >= 2) { cout << "No" << endl; return 0; }
    int u = q.front(); q.pop();
    ++cnt; f[u] = cnt;
    for (int i : vec[u]) {
      --deg[i];
      if (!deg[i]) q.push(i);
    }
  }

  cout << "Yes" << endl;
  for (int i = 1; i <= n; ++i) cout << f[i] << " ";
  cout << endl;
  return 0;
}
```

Happy New Year 2023

先用埃氏筛筛质数, 然后枚举, 枚举的时间复杂度为 3e6

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 3e6 + 5;
bool st[maxn];
vector<int> primes;

void solve() {
  int x; cin >> x;
  for (int i : primes) {
    if (x%(i*i) == 0) {
      cout << i << " " << x/(i*i) << endl;
      break;
    }
    if (x%i == 0) {
      int t = sqrtl(x/i);
      cout << t << " " << i << endl;
      break;
    }
  }
}

signed main()
{
  for (int i = 2; i < maxn; ++i) {
    if (st[i]) continue;
    primes.push_back(i);
    for (int j = i+i; j < maxn; j+=i) st[j] = true;
  }
```

```
    int T; cin >> T;
    while (T -- ) solve();
    return 0;
  }
```